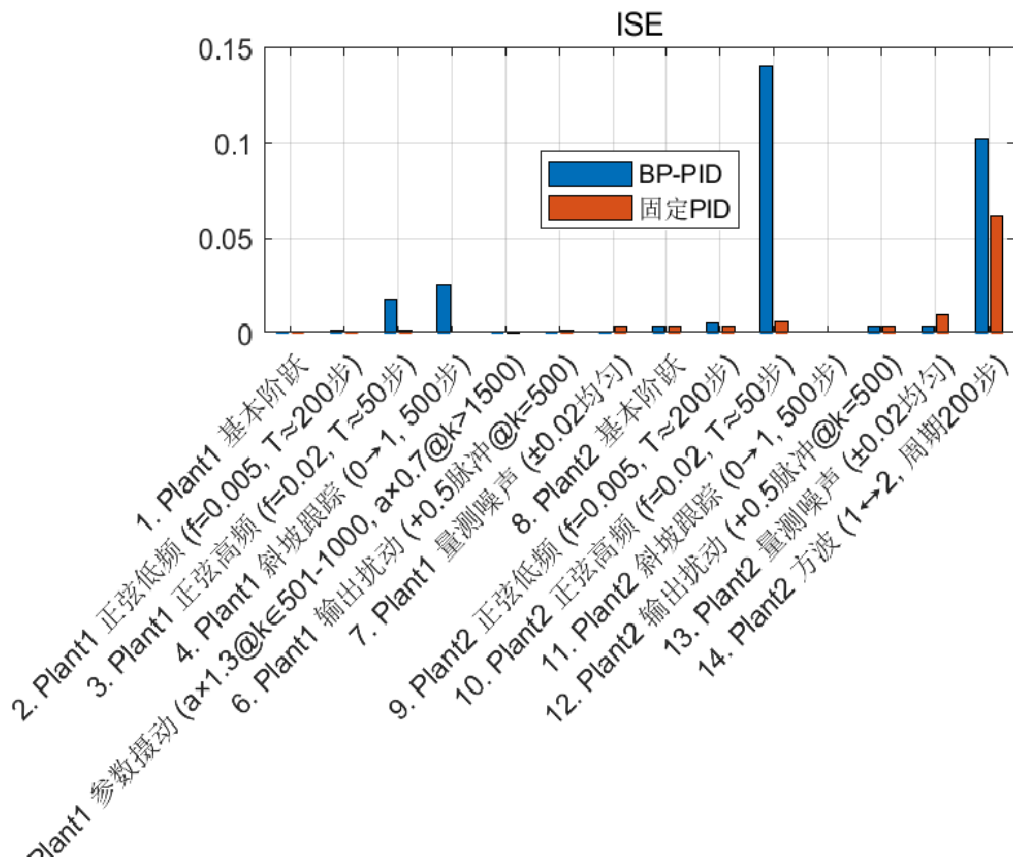
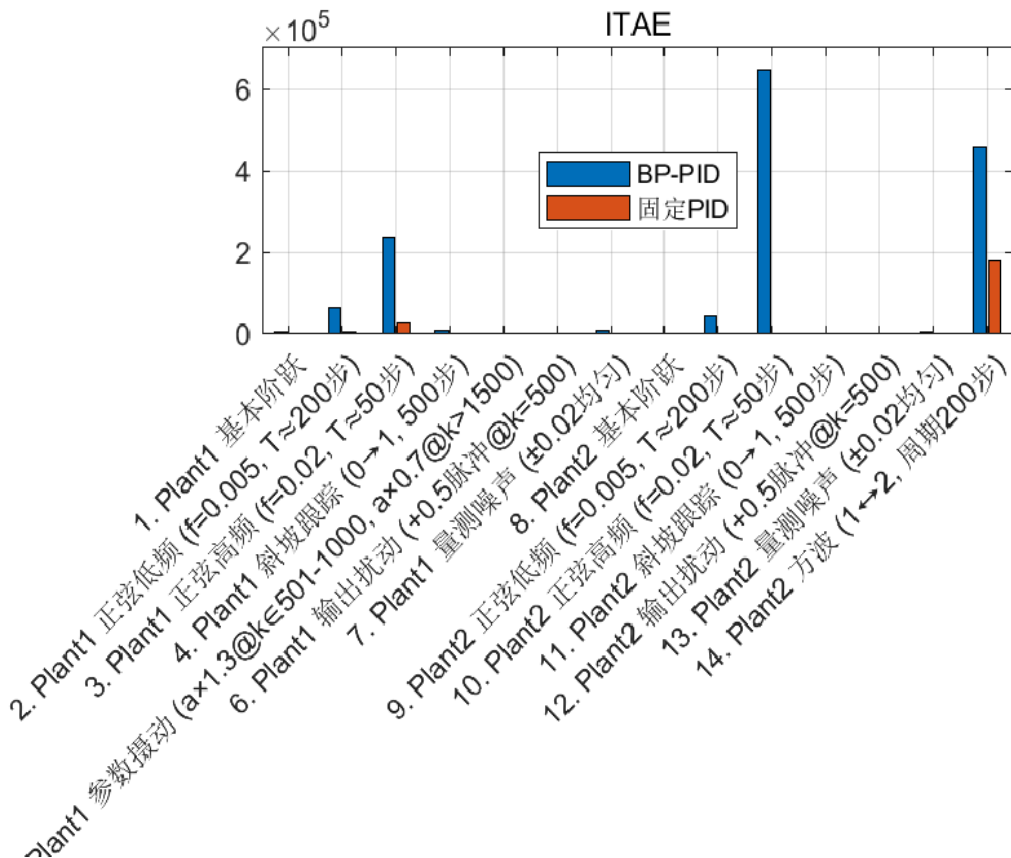
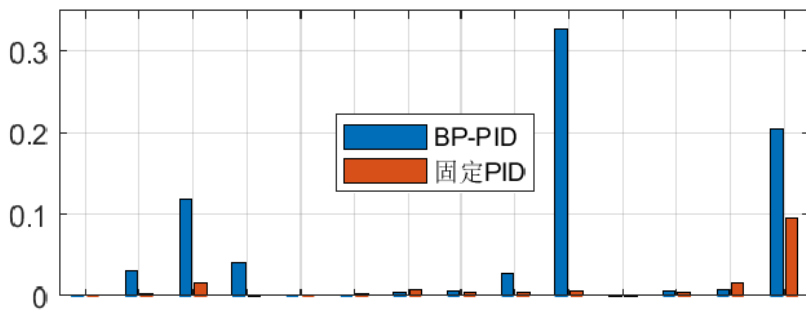




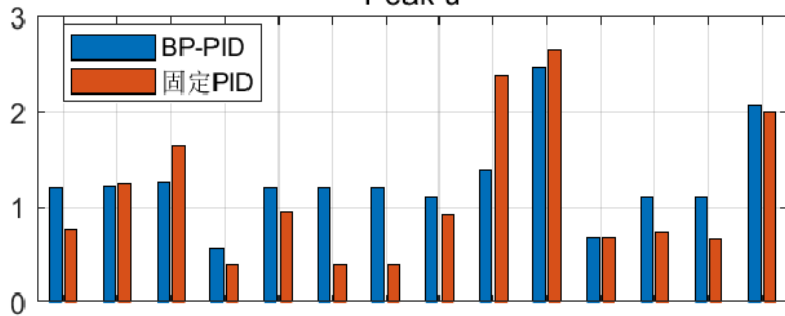


MAE



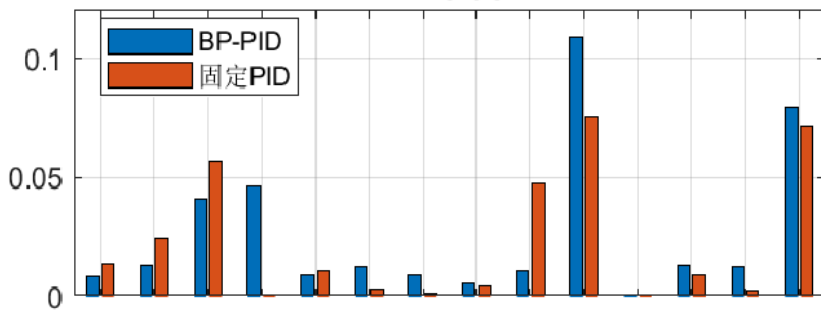
1. Plant1 基本阶跃
 2. Plant1 正弦低频 ($f=0.005$, $T \approx 200$ 步)
 3. Plant1 正弦高频 ($f=0.02$, $T \approx 50$ 步)
 4. Plant1 斜坡跟踪 ($0 \rightarrow 1$, 500步)
 5. Plant1 参数扰动 ($a \times 1.3 @ k \in 501-1000$)
 6. Plant1 输出扰动 ($+0.5$ 脉冲@ $k > 1500$)
 7. Plant1 量测噪声 (± 0.02 均匀)
 8. Plant2 基本阶跃
 9. Plant2 正弦低频 ($f=0.005$, $T \approx 200$ 步)
 10. Plant2 正弦高频 ($f=0.02$, $T \approx 50$ 步)
 11. Plant2 斜坡跟踪 ($0 \rightarrow 1$, 500步)
 12. Plant2 输出扰动 ($+0.5$ 脉冲@ $k \in 501-1000$)
 13. Plant2 量测噪声 (± 0.02 均匀)
 14. Plant2 方波 ($1 \leftrightarrow 2$, 周期200步)

Peak u



1. Plant1 基本阶跃
2. Plant1 正弦低频 ($f=0.005$, $T \approx 200$ 步)
3. Plant1 正弦高频 ($f=0.02$, $T \approx 50$ 步)
4. Plant1 斜坡跟踪 ($0 \rightarrow 1$, 500步)
5. Plant1 量测噪声 ($0 \rightarrow 1$, 500步)
6. Plant1 输出扰动 (± 0.5 脉冲@ $k=500$)
7. Plant1 量测噪声 (± 0.02 均匀)
8. Plant2 基本阶跃
9. Plant2 正弦低频 ($f=0.005$, $T \approx 200$ 步)
10. Plant2 正弦高频 ($f=0.02$, $T \approx 50$ 步)
11. Plant2 斜坡跟踪 ($0 \rightarrow 1$, 500步)
12. Plant2 输出扰动 (± 0.5 脉冲@ $k=500$)
13. Plant2 量测噪声 (± 0.02 均匀)
14. Plant2 方波 ($1 \leftrightarrow 2$, 周期200步)

RMS dU



1. Plant1 基本阶跃

2. Plant1 正弦低频 ($f=0.005$, $T \approx 200$ 步)

3. Plant1 正弦高频 ($f=0.02$, $T \approx 50$ 步)

4. Plant1 斜坡跟踪 ($0 \rightarrow 1$, 500步)

5. Plant1 量测噪声 (± 0.5 脉冲@ $k=500$)

6. Plant1 输出扰动 (± 0.5 脉冲@ $k=500$)

7. Plant1 量测噪声 (± 0.02 均匀)

8. Plant2 基本阶跃

9. Plant2 正弦低频 ($f=0.005$, $T \approx 200$ 步)

10. Plant2 正弦高频 ($f=0.02$, $T \approx 50$ 步)

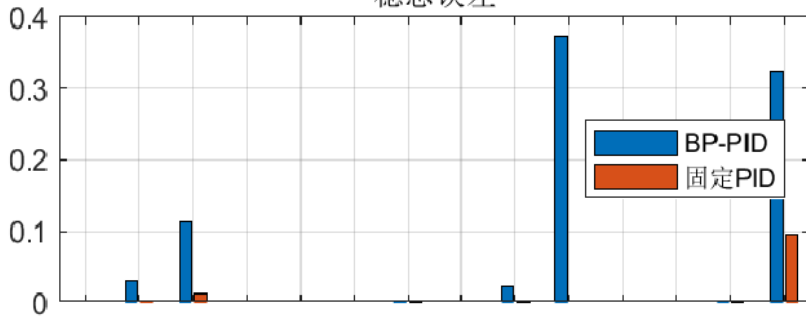
11. Plant2 斜坡跟踪 ($0 \rightarrow 1$, 500步)

12. Plant2 输出扰动 (± 0.5 脉冲@ $k=500$)

13. Plant2 量测噪声 (± 0.02 均匀)

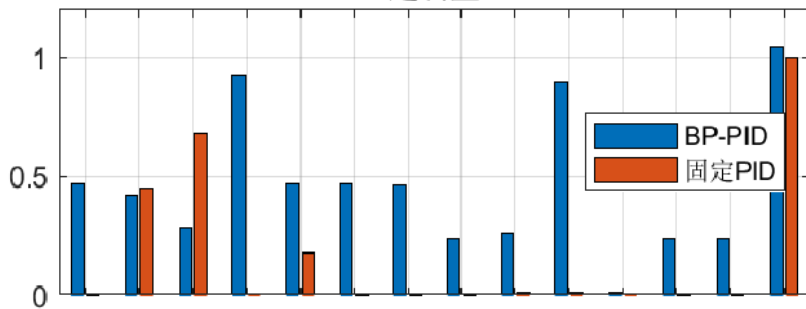
14. Plant2 方波 ($1 \leftrightarrow 2$, 周期200步)

稳态误差



1. Plant1 基本阶跃
2. Plant1 正弦低频 ($f=0.005$, $T \approx 200$ 步)
3. Plant1 正弦高频 ($f=0.02$, $T \approx 50$ 步)
4. Plant1 斜坡跟踪 ($0 \rightarrow 1$, 500步)
5. Plant1 参数摄动 ($a \times 1.3 @ k \in 501-1000$, $a \times 0.7 @ k > 1500$)
6. Plant1 输出扰动 ($+0.5$ 脉冲@ $k=500$)
7. Plant1 量测噪声 (± 0.02 均匀)
8. Plant2 基本阶跃
9. Plant2 正弦低频 ($f=0.005$, $T \approx 200$ 步)
10. Plant2 正弦高频 ($f=0.02$, $T \approx 50$ 步)
11. Plant2 斜坡跟踪 ($0 \rightarrow 1$, 500步)
12. Plant2 输出扰动 ($+0.5$ 脉冲@ $k=500$)
13. Plant2 量测噪声 (± 0.02 均匀)
14. Plant2 方波 ($1 \leftrightarrow 2$, 周期200步)

超调量



1. Plant1 基本阶跃

2. Plant1 正弦低频 ($f=0.005$, $T \approx 200$ 步)

3. Plant1 正弦高频 ($f=0.02$, $T \approx 50$ 步)

4. Plant1 斜坡跟踪 ($0 \rightarrow 1$, 500步)

5. Plant1 量测噪声 ($0 \rightarrow 1$, 500步)

6. Plant1 输出扰动 ($+0.5$ 脉冲@ $k=500$)

7. Plant1 量测噪声 (± 0.02 均匀)

8. Plant2 基本阶跃

9. Plant2 正弦低频 ($f=0.005$, $T \approx 200$ 步)

10. Plant2 正弦高频 ($f=0.02$, $T \approx 50$ 步)

11. Plant2 斜坡跟踪 ($0 \rightarrow 1$, 500步)

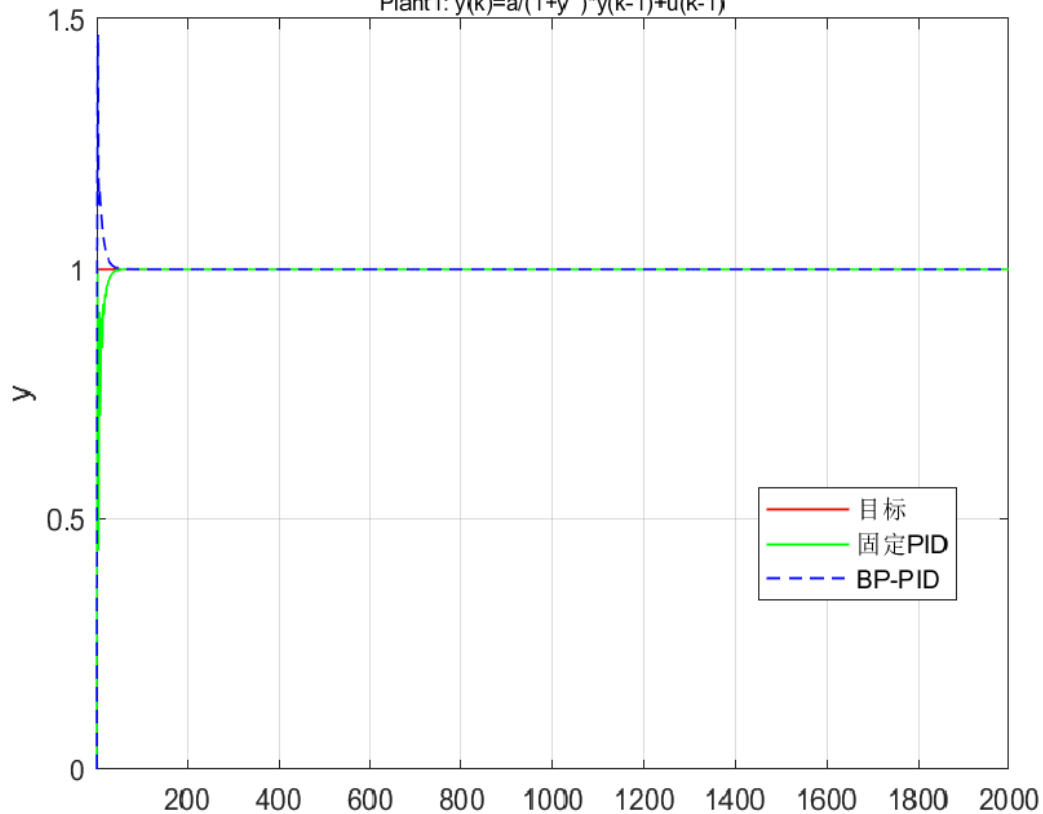
12. Plant2 输出扰动 ($+0.5$ 脉冲@ $k=500$)

13. Plant2 量测噪声 (± 0.02 均匀)

14. Plant2 方波 ($1 \leftrightarrow 2$, 周期200步)

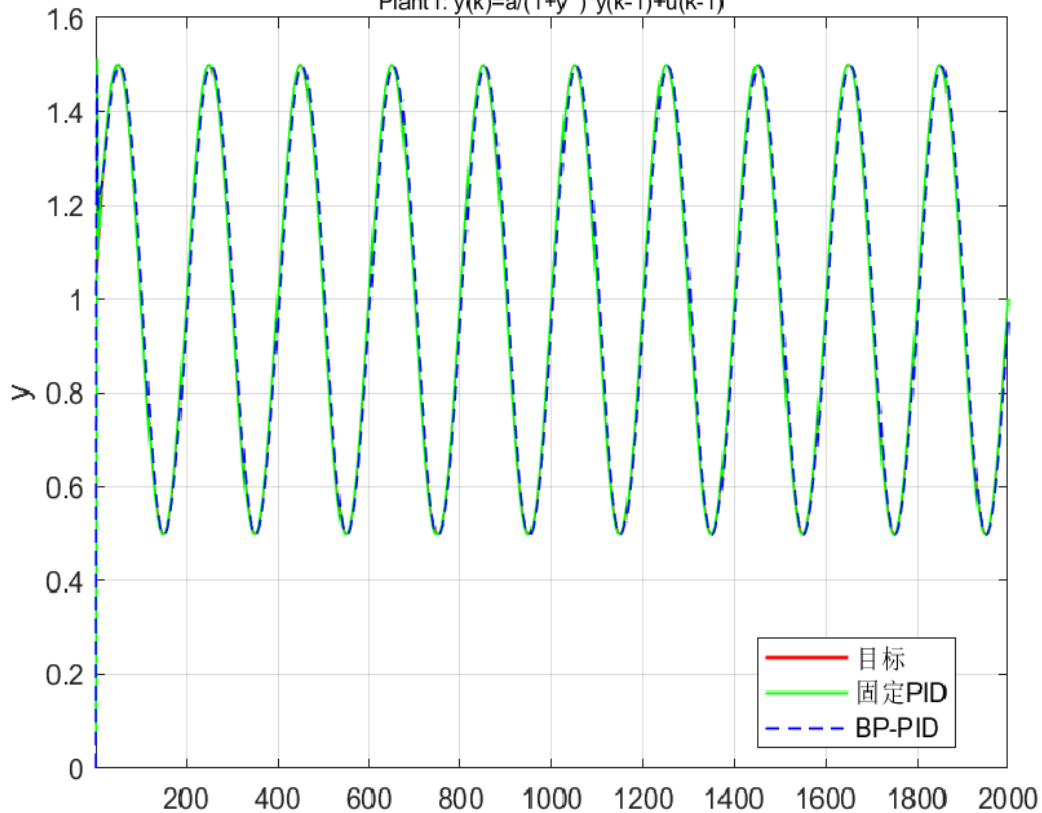
1. Plant1 基本阶跃

$$\text{Plant1: } y(k) = a/(1+y^2) * y(k-1) + u(k-1)$$



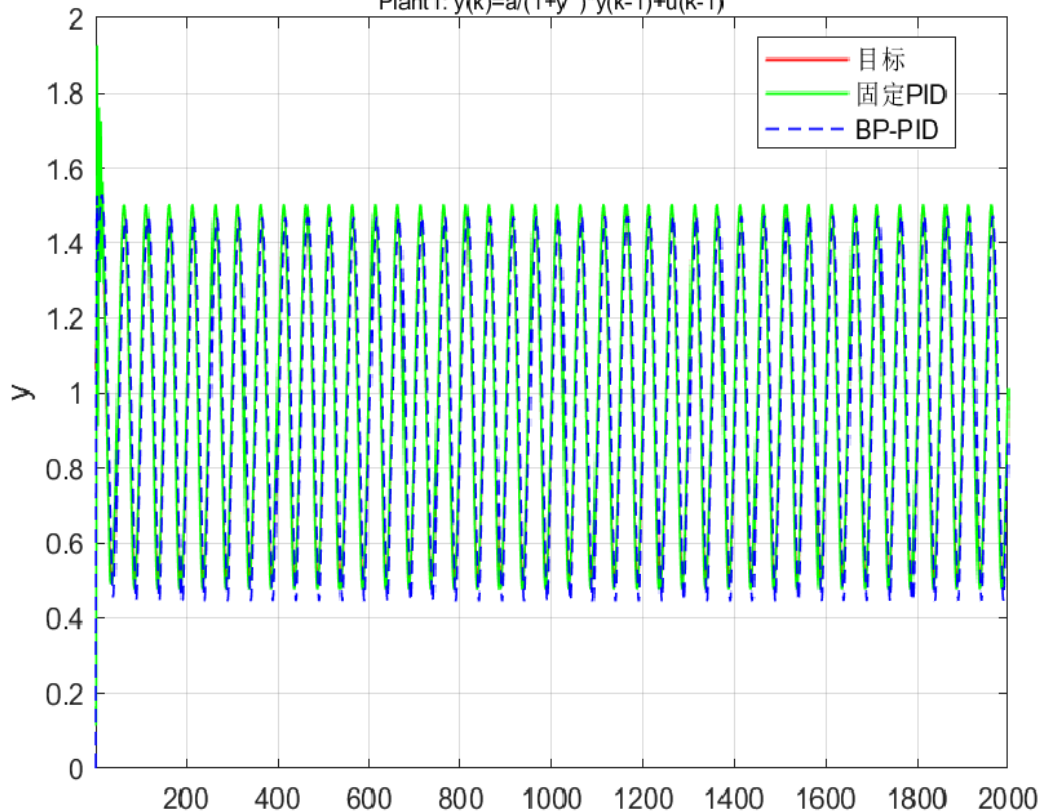
2. Plant1 正弦低频 ($f=0.005$, $T \approx 200$ 步)

Plant1: $y(k)=a/(1+y^2)*y(k-1)+u(k-1)$



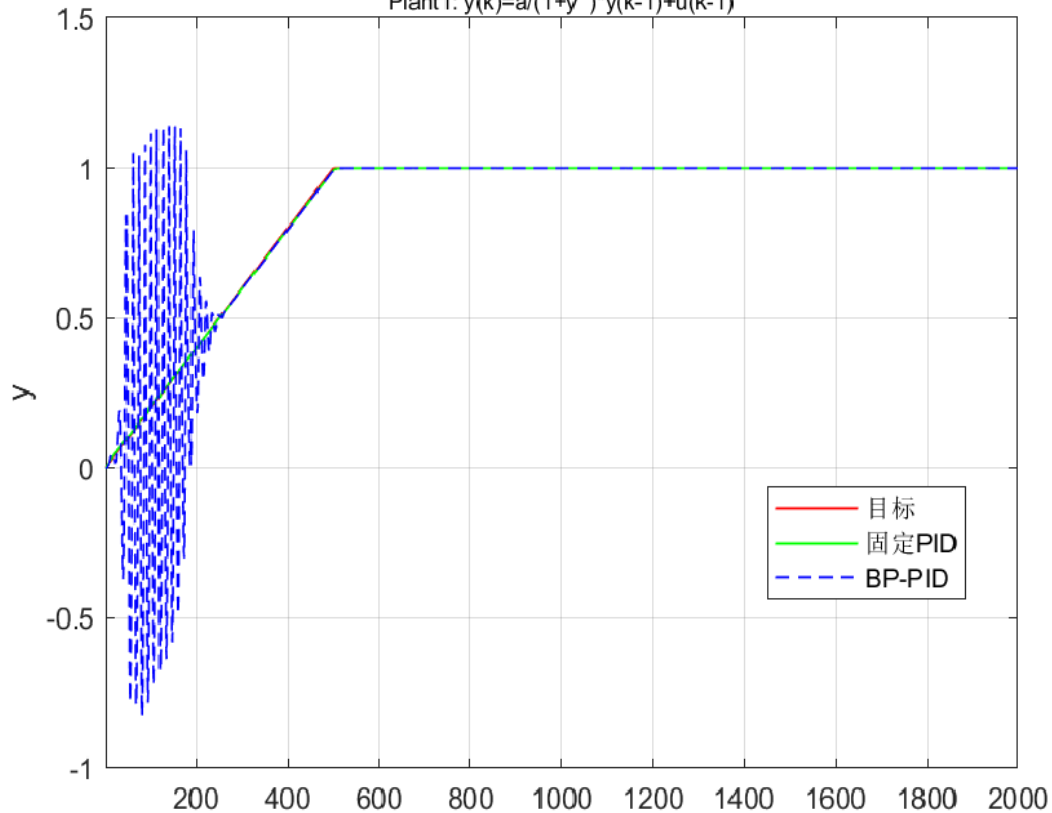
3. Plant1 正弦高频 ($f=0.02$, $T \approx 50$ 步)

$$\text{Plant1: } y(k) = a/(1+y^2) * y(k-1) + u(k-1)$$



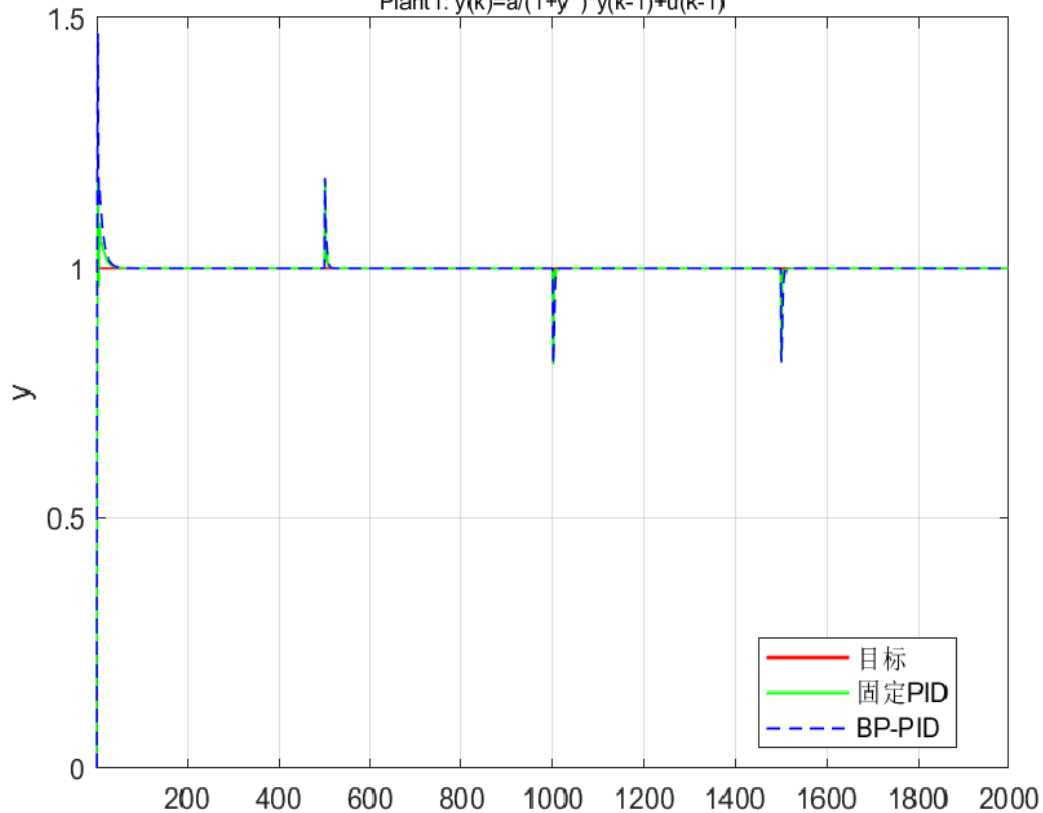
4. Plant1 斜坡跟踪 (0→1, 500步)

Plant1: $y(k) = a/(1+y^2) * y(k-1) + u(k-1)$



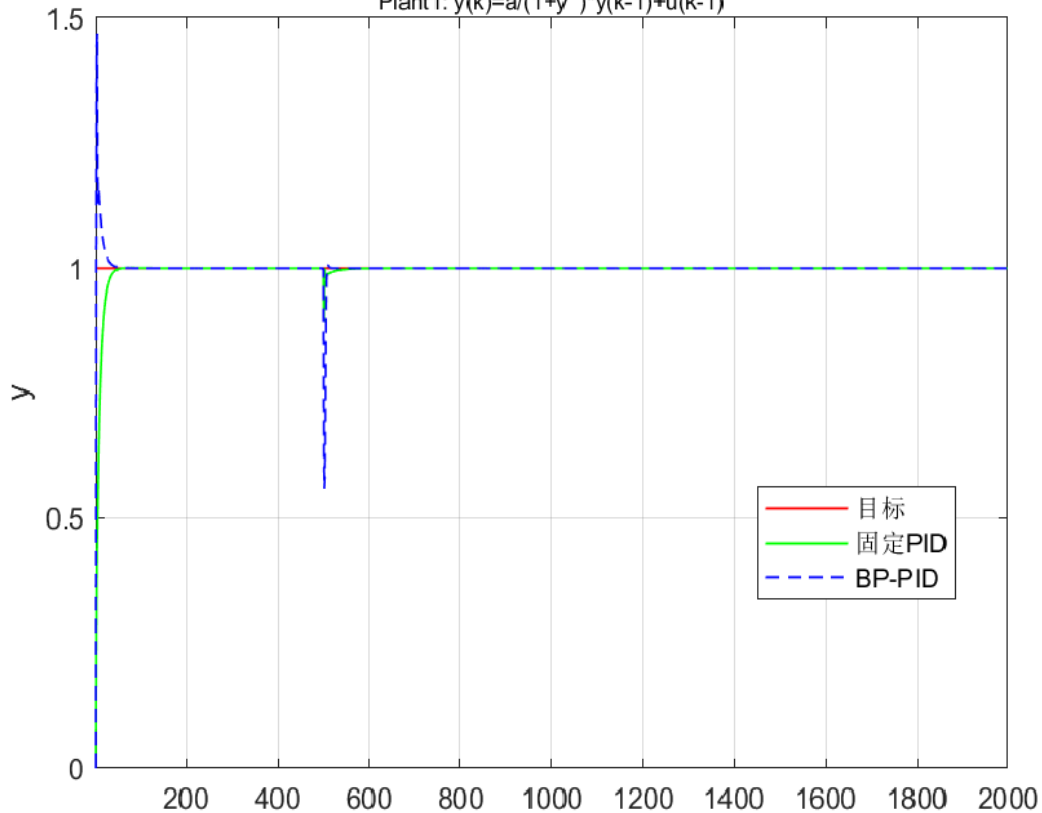
5. Plant1 参数摄动 ($a \times 1.3 @ k \in 501-1000$, $a \times 0.7 @ k > 1500$)

$$\text{Plant1: } y(k) = a/(1+y^2) * y(k-1) + u(k-1)$$



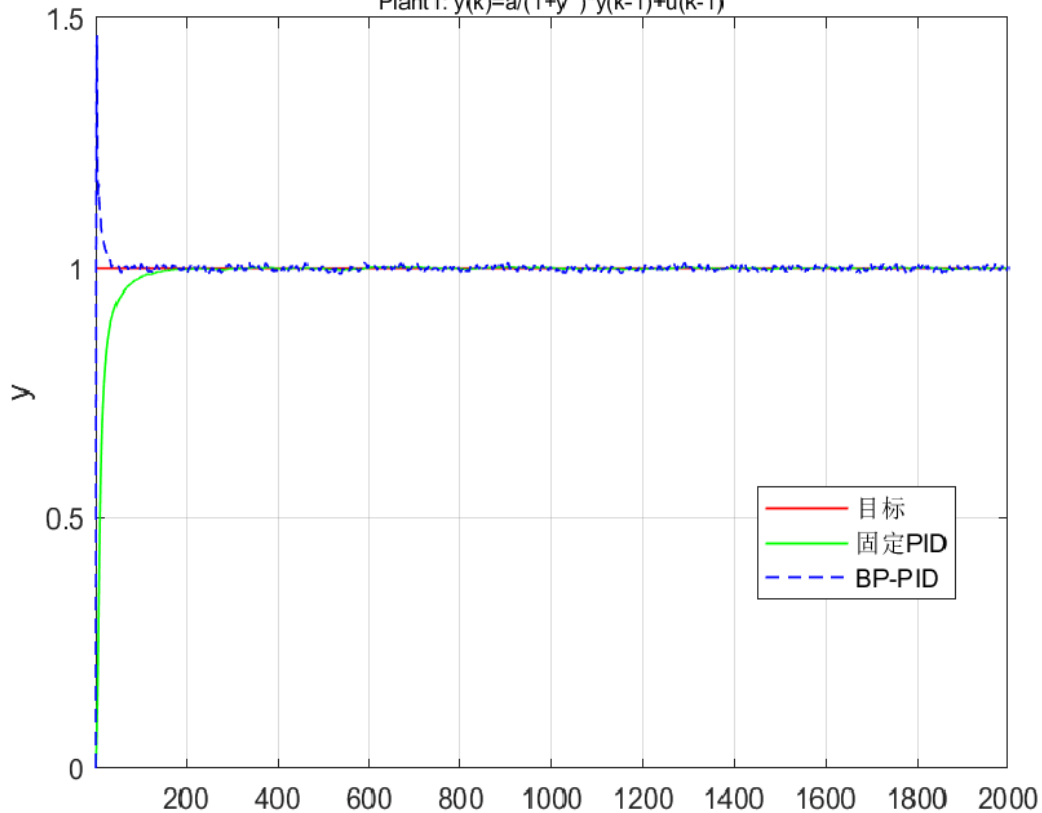
6. Plant1 输出扰动 (+0.5脉冲@k=500)

$$\text{Plant1: } y(k) = a/(1+y^2) * y(k-1) + u(k-1)$$



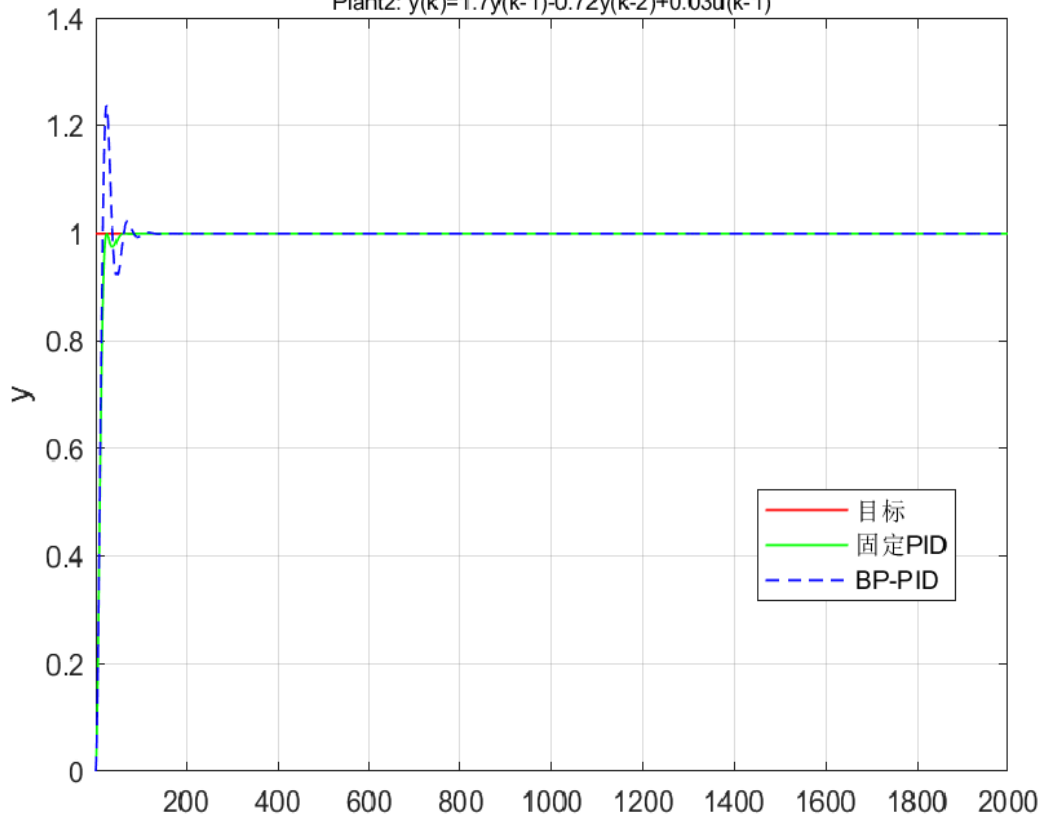
7. Plant1 量测噪声 (± 0.02 均匀)

$$\text{Plant1: } y(k) = a/(1+y^2) * y(k-1) + u(k-1)$$



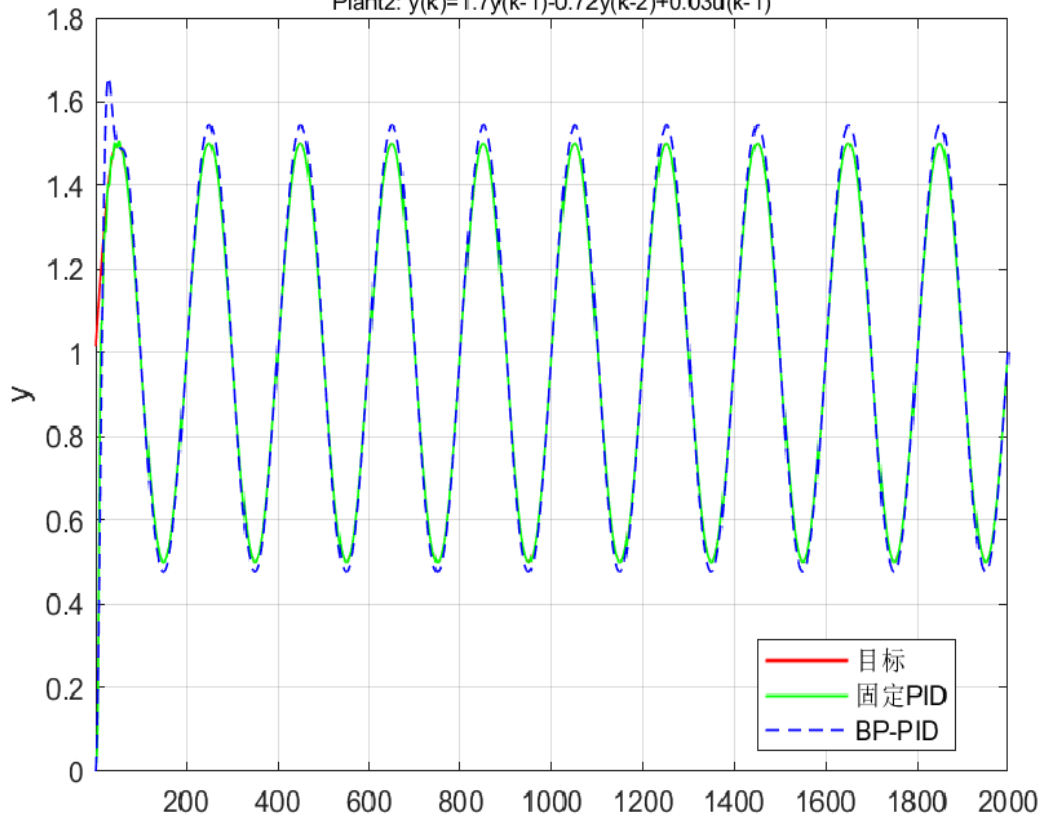
8. Plant2 基本阶跃

$$\text{Plant2: } y(k) = 1.7y(k-1) - 0.72y(k-2) + 0.03u(k-1)$$



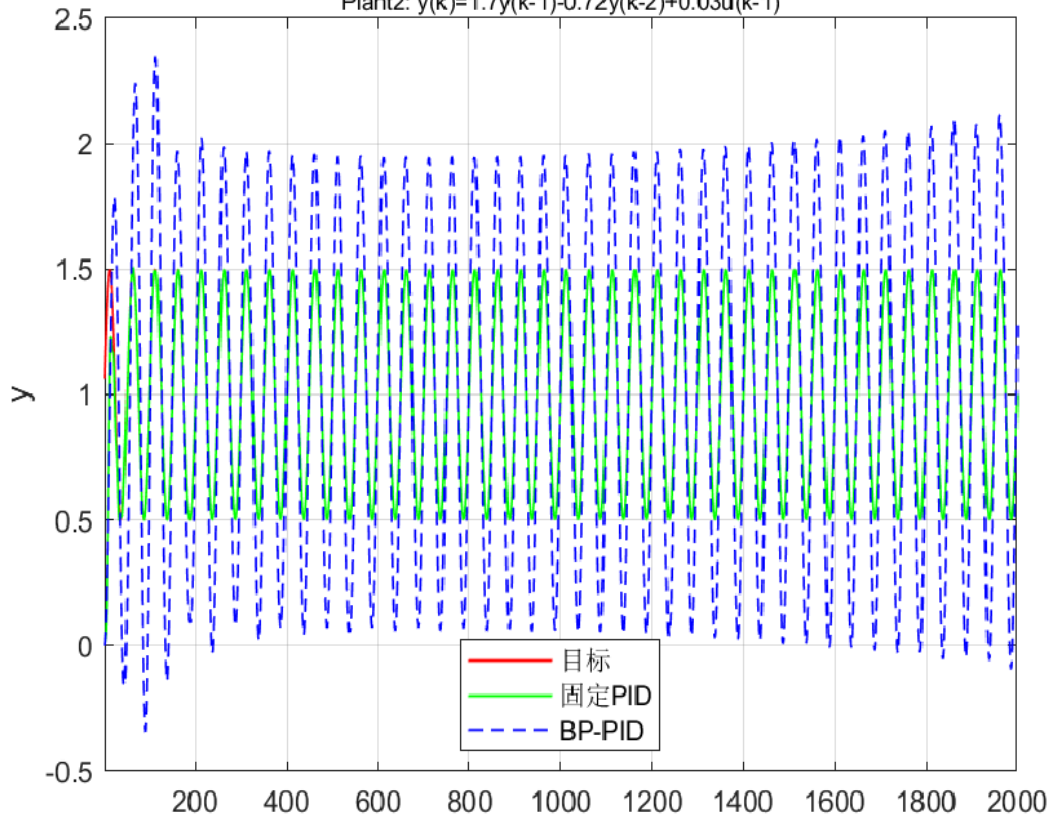
9. Plant2 正弦低频 ($f=0.005$, $T \approx 200$ 步)

$$\text{Plant2: } y(k) = 1.7y(k-1) - 0.72y(k-2) + 0.03u(k-1)$$



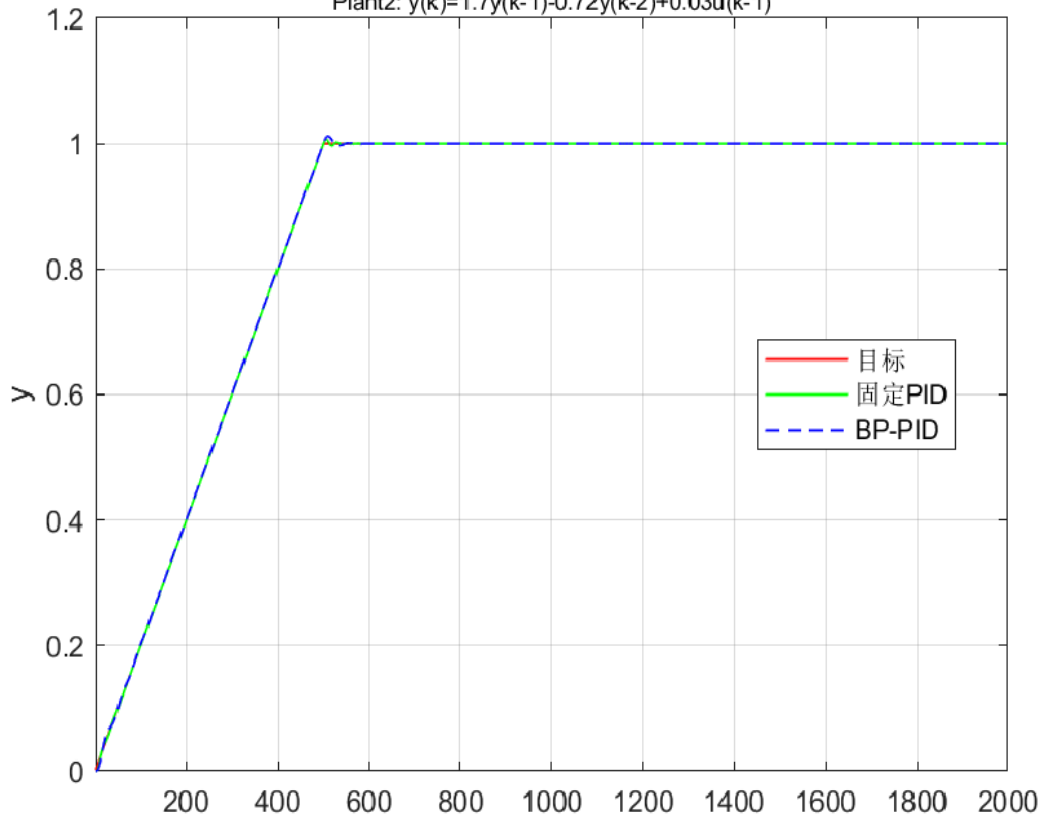
10. Plant2 正弦高频 ($f=0.02$, $T \approx 50$ 步)

$$\text{Plant2: } y(k) = 1.7y(k-1) - 0.72y(k-2) + 0.03u(k-1)$$



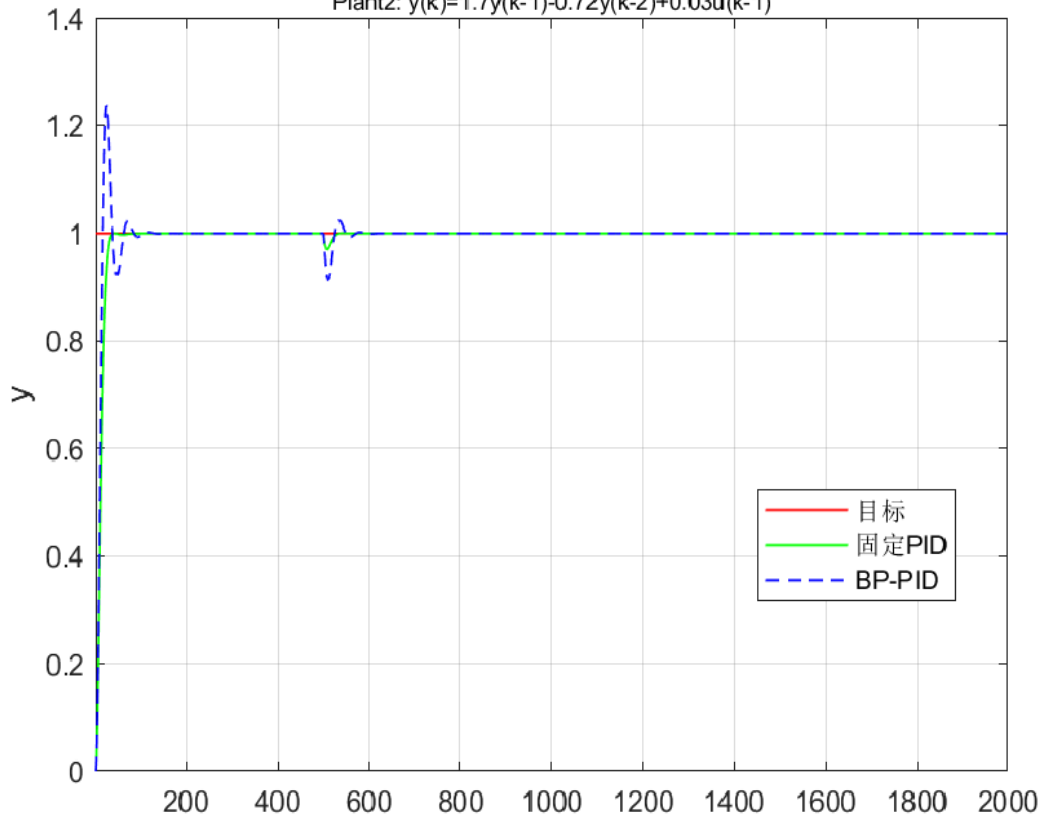
11. Plant2 斜坡跟踪 (0→1, 500步)

Plant2: $y(k)=1.7y(k-1)-0.72y(k-2)+0.03u(k-1)$

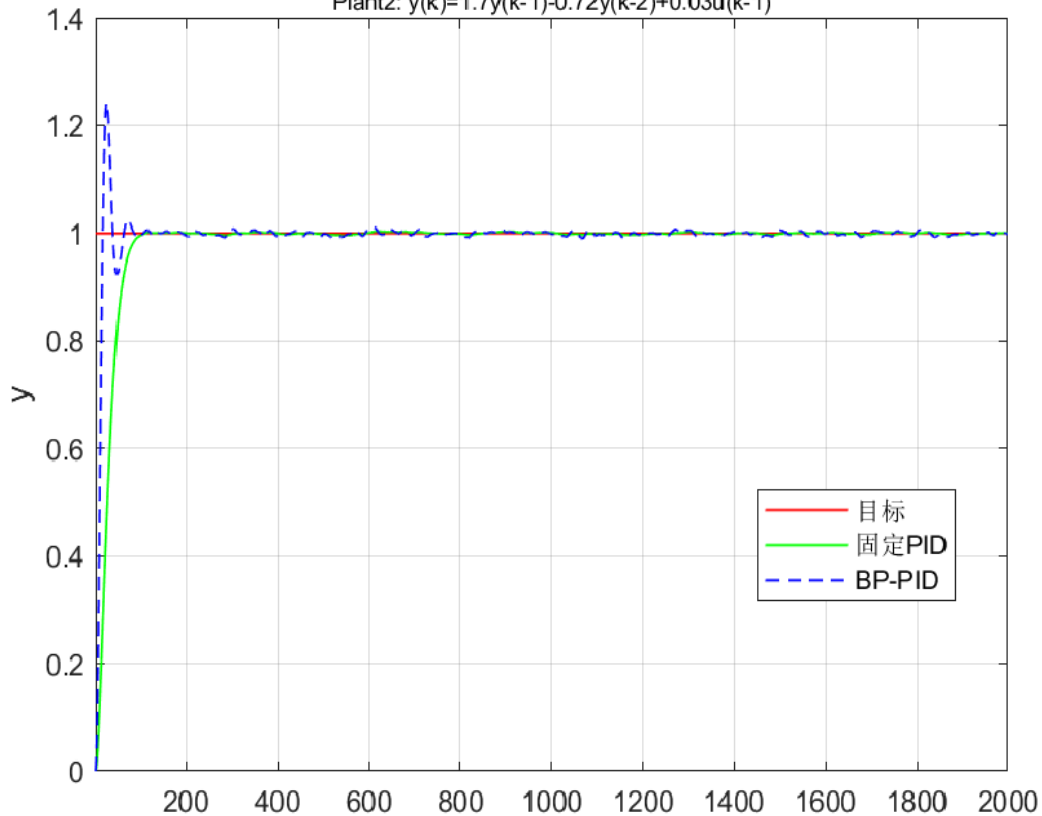


12. Plant2 输出扰动 (+0.5脉冲@k=500)

Plant2: $y(k)=1.7y(k-1)-0.72y(k-2)+0.03u(k-1)$



13. Plant2 量测噪声 (± 0.02 均匀)
Plant2: $y(k) = 1.7y(k-1) - 0.72y(k-2) + 0.03u(k-1)$



14. Plant2 方波 (1↔2, 周期200步)

$$\text{Plant2: } y(k) = 1.7y(k-1) - 0.72y(k-2) + 0.03u(k-1)$$

